PR #988 measurement qualification

Online-frame experiment | run 39303742695

This note supplements the frozen report for `koblitz-n83-onlineframe-20261019`, run `39303742695`. It preserves the run measurements and its frozen-protocol verdict.

Timing interpretation

The archived primary compares one cS and one cF timing per point across 64 points, with arm order randomized within each point. The cF/cFaa repetitions are a separate A/A control; they are not repeated alternating cS/cF primary pairs. The frozen protocol classifies the online-stage result as an improvement, with ratio **1.5733 [1.5714, 1.5753]**. The repository current Tier 1 speedup gate requires repeated alternating pairs and a 95% paired confidence interval whose lower bound exceeds 1. This single primary pair per point does not meet the repeated-pair requirement. Treat the estimate as a historical, instance-scoped online-stage result, not a qualified Tier 1 speedup claim. The independent end-to-end secondary is **1.0008 [1.0006, 1.0010]** with shared rank-stage cost; the experiment reports that result as a pipeline improvement under its frozen rule, but it also has only the same one paired observation per point.

Memory and accounting fields

The report per-arm peak RSS values come from the producer getrusage(RUSAGE_SELF) field peak_rss_bytes, so the reported per-process peak-RSS comparison is available. The runner also records child_peak_rss_kb_cumulative from RUSAGE_CHILDREN.ru_maxrss; that field is a cumulative child-process high-water mark, not a per-row peak. Runner cgroup counters are allocation-level. Process user/system CPU-time and wall-time observations remain in the archived per-run rows. These distinctions do not change the recorded timing estimate or the frozen protocol checks.

Policy and evidence

Policy reference: current AGENTS.md, CPU performance isolation gate, Tier 1 paragraph (repeated alternating pairs; paired 95% lower bound above 1). This is a current reporting qualification; it does not change the historical experiment frozen rule.

Validation: derived from exact PR head 6c59e4da048ce9a64204a1fafc73ee681175ea6c, main base 517073ab2dfed83584ddc1115721b7c251620e5b; full archived receipt replay passed with pinned sources. See the PR body for source and replay evidence.